

Two-Robot Intelligent Warehouse Navigation System Using Advanced Graph Algorithms

The rapid growth of warehouse automation has increased the demand for intelligent robotic systems capable of navigating efficiently, avoiding collisions, and adapting to dynamic environments. This project proposes a **Two-Robot Intelligent Warehouse Navigation System** that applies advanced graph algorithms to solve real-world warehouse management challenges using physical mobile robots.

In the proposed system, the warehouse layout is modeled as a **graph**, where intersections and checkpoints are represented as **nodes** and the connecting pathways as **edges**. Each robot is assigned a source and destination location, and the optimal route is computed using shortest path algorithms such as **A*** or **Dijkstra's Algorithm**. To support multiple robots operating simultaneously, a time-based scheduling mechanism is introduced to detect and prevent collisions when two robots attempt to occupy the same node at the same time. Collision resolution is achieved through priority-based path replanning and dynamic route allocation.

The system is further enhanced with **dynamic obstacle handling**, where obstacles detected by onboard ultrasonic sensors cause the affected edges to be temporarily removed from the graph or assigned infinite cost values. The robots then recompute alternative routes without requiring reconstruction of the entire graph, enabling real-time adaptation to changing warehouse conditions.

To improve localization accuracy, **RFID tags** are placed at every node in the warehouse, while each robot carries an RFID reader to identify its current position during navigation. Additionally, graph-based task allocation techniques such as **bipartite matching** can be incorporated to optimize the assignment of delivery tasks among multiple robots, minimizing travel distance and execution time. An optional communication layer based on **Minimum Spanning Tree (MST)** principles can also be used to reduce communication cost and power consumption among robots and the central controller.

The proposed system demonstrates a practical hardware implementation of advanced graph algorithms in robotics and warehouse automation. By combining shortest path computation, collision avoidance, dynamic graph updates, and intelligent task scheduling, the project provides a scalable foundation for future smart warehouse and industrial automation systems.